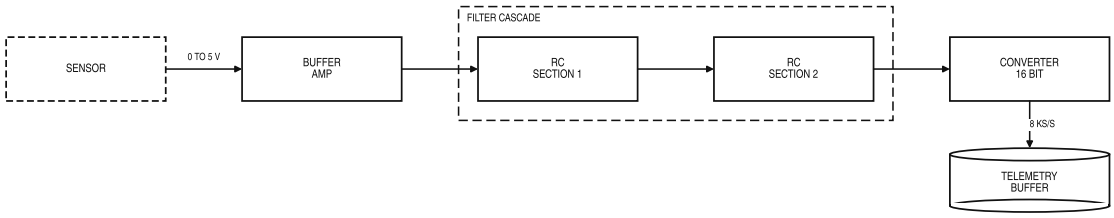
# ANTI-ALIAS FILTER

*an engineering section with a block diagram, math, a log-frequency plot and a parts table*

## 4.2.1 FUNCTIONAL DESCRIPTION.

Each sensor channel passes through a cascade of first-order RC sections ahead of the converter. The cascade removes energy above the Nyquist frequency so that it cannot fold back into the measurement band.




SIG-501A

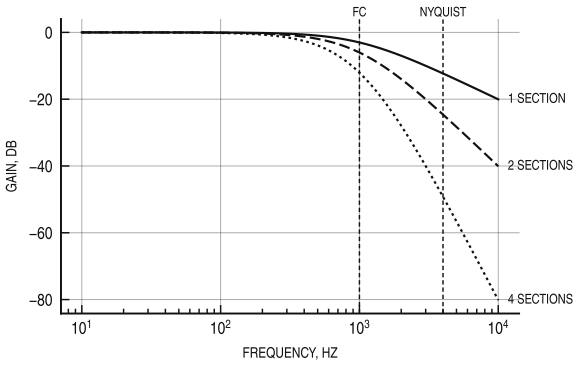
Figure 4.2-1. Signal Conditioning Channel, One of Eight

### 4.2.1.1 Transfer function.

One section passes a fraction of the input set by the ratio of frequency to the corner frequency $f_c$:

$$H(f) = \frac{1}{1 + jf/f_c}, \qquad f_c = \frac{1}{2\pi RC}$$

With $R = 1.6\ \text{k}\Omega$ and $C = 100\ \text{nF}$ the corner is $f_c \approx 1.0\ \text{kHz}$. Each section adds 6 dB per octave of rolloff above the corner (figure 4.2-2).




SIG-502A

Figure 4.2-2. Magnitude Response of One, Two and Four Sections

Table 4.2-1. Section Parts

| Ref | Value | Tol | Note |
|---|---|---|---|
| R1, R2 | 1.6 k | 1 % | metal film |
| C1, C2 | 100 nF | 5 % | C0G |
| U1 | -- | -- | rail-to-rail buffer |

**CAUTION**

Do not substitute X7R capacitors. Their value falls with applied voltage and moves the corner.

## 4.2.2 VERIFICATION.

```
CHANNEL RESPONSE CHECK                                  TEST
SIG GEN ............... 1 KHZ, 1 V RMS
CHAN 1 OUTPUT ......... -3 DB +/- 0.5
SIG GEN ............... 4 KHZ
CHAN 1 OUTPUT ......... -24 DB OR LESS
REPEAT ................ CHAN 2 THRU 8
```